

ASOSAN, Japan

WMO: 478210

Lat: 32.8804N

Lon: 131.0733E

Elev: 3755

StdP: 12.81

Time Zone: 9.00 (E09)

Period: 94-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.7	18.1	-1.9	5.7	37.2	4.8	8.1	36.1	27.7	32.1	25.1	30.8	14.2	320	0.567

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.4	78.4	69.8	76.5	68.9	74.6	67.9	71.4	75.6	70.3	73.9	69.5	72.5	8.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	127.9	73.0	69.3	123.9	71.8	68.5	120.7	70.9	37.9	75.8	37.0	74.2	36.1	72.9	76.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.2	23.5	20.7	DB	11.8	81.3	2.8	1.7	9.8	82.5	8.1	83.5	6.5	84.4	4.4	85.6	(4)
			WB	11.3	73.4	2.9	1.2	9.2	74.3	7.5	75.0	5.9	75.7	3.8	76.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	29.4	32.5	38.6	48.1	57.0	62.6	69.0	70.2	64.7	55.2	44.6	33.8	(6)
		DBStd	15.25	7.03	7.96	7.44	6.56	4.49	3.46	3.10	2.64	4.58	5.54	7.18	7.47	(7)
		HDD50	2310	638	490	356	109	5	0	0	0	19	190	503		(8)
		HDD65	5625	1103	910	817	507	250	87	5	2	60	305	611	968	(9)
		CDD50	2521	0	1	4	51	221	377	590	627	442	179	29	0	(10)
		CDD65	362	0	0	0	0	2	14	130	163	52	1	0	0	(11)
		CDH74	522	0	0	0	0	4	5	220	257	36	0	0	0	(12)
		CDH80	15	0	0	0	0	0	0	8	7	0	0	0	0	(13)
		WSAvg	10.2	10.7	10.7	11.2	10.9	10.1	10.3	10.5	9.6	9.5	9.3	9.1	10.2	(14)
		PrecAvg	108.3	3.2	4.5	7.0	8.0	9.5	23.1	19.8	11.1	10.6	5.0	4.2	3.0	(15)
(16)	Precipitation	PrecMax	191.1	6.6	8.9	18.6	14.6	17.1	49.8	43.9	28.6	28.3	11.7	9.6	5.9	(16)
(17)		PrecMin	61.6	0.1	1.5	2.9	2.9	3.2	7.3	4.3	2.5	1.9	0.4	0.9	0.2	(17)
(18)		PrecStd	24.3	1.6	2.0	3.2	3.2	3.9	10.2	11.2	6.2	6.8	3.0	2.2	1.5	(18)
(19)		PrecStd	24.3	1.6	2.0	3.2	3.2	3.9	10.2	11.2	6.2	6.8	3.0	2.2	1.5	(18)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	50.9	55.3	58.8	67.5	73.7	74.7	81.1	80.8	77.5	70.4	62.3	53.4		(19)
(21)		MCWB	47.9	50.1	47.9	54.7	56.6	64.2	70.1	70.8	68.4	60.9	55.3	50.2		(20)
(22)		DB	47.0	51.7	55.2	63.0	69.8	71.9	78.6	78.7	74.7	66.9	59.5	49.9		(21)
(23)		MCWB	43.7	48.5	48.2	52.1	55.6	62.5	69.8	70.2	66.9	59.4	54.9	45.8		(22)
(24)		DB	43.5	48.6	52.3	60.2	66.9	69.7	76.5	76.9	72.6	64.8	57.4	47.3		(23)
(25)		MCWB	40.9	45.6	47.0	52.3	55.3	62.7	69.2	69.6	65.9	59.2	53.4	43.7		(24)
(26)		DB	40.0	44.9	49.8	57.7	64.5	67.8	74.3	75.1	70.4	63.0	54.9	44.9		(25)
(27)		MCWB	36.6	41.3	45.4	50.7	55.7	63.7	68.5	68.8	65.3	58.3	51.3	41.0		(26)
(28)		WB	50.1	53.1	54.9	61.2	64.8	69.3	72.9	72.9	70.8	65.5	60.0	51.9		(27)
(29)		MCDB	50.4	53.4	55.3	62.4	65.6	70.9	77.5	77.9	74.8	67.3	60.4	52.1		(28)
(30)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	45.5	49.8	51.8	58.4	62.9	67.6	71.3	71.7	69.1	63.6	57.2	48.0		(29)
(31)		MCDB	46.0	50.4	52.7	59.8	64.1	68.5	75.5	76.0	71.8	65.2	58.0	48.6		(30)
(32)		WB	41.5	46.5	49.3	55.7	61.0	66.7	70.3	70.7	68.0	61.8	54.6	45.1		(31)
(33)		MCDB	42.6	48.3	50.7	57.5	62.7	67.5	73.8	74.4	70.3	63.5	56.2	46.3		(32)
(34)		WB	37.4	42.6	46.5	53.1	59.1	65.7	69.4	69.8	67.0	59.7	52.3	42.1		(33)
(35)		MCDB	39.0	44.3	48.4	55.5	61.3	66.6	72.4	73.1	69.2	61.6	54.2	44.0		(34)
(36)	Mean Daily Temperature Range	MDBR	10.2	12.2	13.4	13.6	12.9	9.3	8.7	9.4	10.1	11.5	11.3	10.7		(35)
(37)		5% DB	13.4	15.1	15.2	15.9	16.3	12.6	12.7	12.2	11.7	12.1	12.1	12.9		(36)
(38)		MCWBR	13.2	13.6	12.6	12.1	11.2	8.4	7.7	7.4	7.5	9.1	10.1	12.0		(37)
(39)		5% WB	12.7	13.6	13.4	11.6	10.7	7.4	10.6	10.6	9.1	9.2	10.1	11.6		(38)
(40)		MCWBR	13.6	14.0	13.7	10.7	10.0	6.2	6.9	6.7	6.3	7.5	9.7	12.3		(39)
(41)		MCWBR	13.6	14.0	13.7	10.7	10.0	6.2	6.9	6.7	6.3	7.5	9.7	12.3		(39)
(42)	Clear-Sky Solar Irradiance	taub	0.342	0.373	0.419	0.435	0.443	0.453	0.458	0.440	0.414	0.376	0.348	0.336		(40)
(43)		taud	2.304	2.242	2.093	2.104	2.142	2.196	2.232	2.281	2.319	2.350	2.370	2.341		(41)
(44)		Ebn at Noon	274	276	271	272	270	266	264	267	270	272	270	269		(42)
(45)		Edh at Noon	33	39	49	50	49	46	44	42	39	35	31	30		(43)
(44)	All-Sky Solar Radiation	RadAvg	779	979	1296	1522	1612	1321	1545	1597	1311	1113	854	714		(44)
(45)		RadStd	71	87	83	148	184	197	209	153	146	121	85	50		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page